

Hamro Hospital

Clean HIS Workflow Integration Report

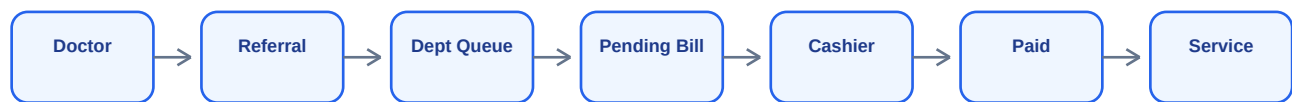
Doctor, Referral, Department, Billing, Cash Counter, Admission, Nursing, OT, Blood Bank, Laboratory, Radiology, Pharmacy and Finance workflow with clear remaining recommendations.

Important clarification

The latest implementation did not intentionally redesign the main frontend or backend UI. Existing dashboard cards, sidebars, tables, buttons, and Jazzmin admin layout were kept. New items were added only where required for workflow. Print and receipt documents are the only area where visual/professional formatting was intentionally improved because hospital logo, patient barcode, bill barcode, and clear print identity were required.

This clean PDF replaces the previous report and fixes broken table/text layout so it is easier to read.

2. Current Project Connection Summary



The system now has a practical connected workflow. A doctor searches a patient, creates a referral or order, the destination department receives it, a pending bill can be generated, Cash Counter collects payment, payment status changes to Paid, and the department proceeds.

Module	Connection now present
Doctor	Search patient, view patient history, create referrals, view own referrals, update profile.
Referral	Stores patient, destination, diagnosis, reason, notes, instructions, attachments, requested items and related bill.
Notification	Sends role or doctor notifications with link to referral detail.
Billing	Supports Pending and Paid status, bill barcode, patient barcode, and selected-bill payment.
Cash Counter	Shows active pending bills and lets cashier pay selected bills.
Admission	Receives admission referrals, has ward daily rate, charge estimate, and pending-bill discharge block.
Patient Record	Doctors can open patient sections and view history read-only.

3. UI and UX Protection

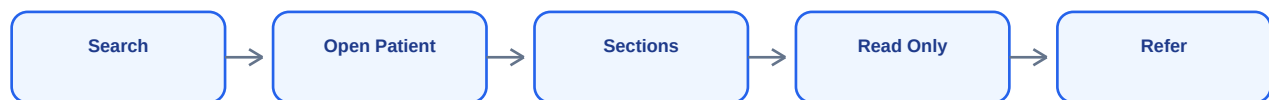
Your instruction is to keep the existing UI clean and stable. The current logic follows that rule. Functional blocks were inserted into existing page styles; no global theme redesign was done in this HIS workflow step.

Area	Status	Explanation
Main dashboards	Kept	Existing dashboard_base, cards and tables remain.
Admin/Jazzmin	Kept	Only backend save/link logic was repaired.
Doctor pages	Feature added	Search, referral and profile features added with existing style.
Department pages	Feature added	Incoming referral panels added using existing card/table style.
Receipts and print	Improved	Print identity was required: logo, barcodes, status, patient info.

Rule going forward

Lock the UI. Add only required functional sections. For any printed document, use one hospital print identity with logo, patient barcode, document barcode, amount/status, printed by, and printed date/time.

4. Doctor View of Patient Records



Doctors can search by Hospital ID, barcode text, phone number or name. After opening the patient, they can access history sections from the patient profile hub.

Patient area	Doctor permission
Profile and demographics	View
Patient card and barcode	View only
Billing and receipts	View
Laboratory and radiology reports	View
Pharmacy, admission and blood bank history	View
Medical documents	View only
Document download	No, only Super Admin and Registration Counter
Patient card print/download	No, only Super Admin and Registration Counter

5. Doctor Account and Profile Logic

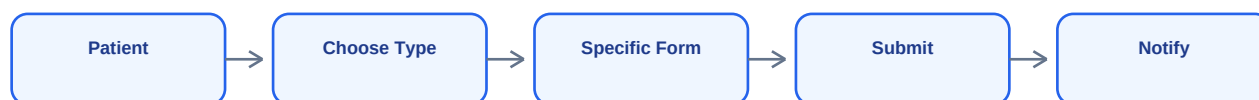
Doctor profiles are linked with hospital staff user accounts. When a doctor is created and no user is linked, the system creates a Doctor-role staff account and attaches it to that doctor profile.

Issue	Logic added
Doctor cannot login	Auto-create or link User account with Doctor role.
Doctor dashboard incomplete	Dashboard, search, referrals, profile and password pages added.
Profile not linked warning	Active doctor profiles are linked automatically when created.
eSewa warning	eSewa fields are optional; no forced warning when blank.
Doctor becomes patient	Doctor user and patient record remain separate.

Recommended improvement

Add a dedicated staff onboarding page with generated credentials, first-login password reset, and optional SMS/email delivery.

6. Referral Category Workflow



Doctors now choose the referral category first. This prevents one very long form and makes each department request clearer.

Referral type	Captured information
Laboratory	LabTest catalogue checkbox selection plus notes and attachment.
Radiology	RadiologyTest checkbox selection plus notes and attachment.
Operation Theatre	OperationType/procedure choices plus notes.
Blood Bank	Blood group, units, urgency, crossmatch and product details.
Admission	Admission notes, reason and expected ward/care information.
Nursing	Nursing instructions such as IV fluid, dressing and monitoring.
Pharmacy	Requested medicines/items and attachment.
Other Doctor/Department	General referral to destination department doctors.

7. Notification and Queue Logic

When a referral is created, the destination receives a notification. The notification contains a related link so staff can open the referral directly.

Destination	Notification target
Laboratory	Laboratory role users
Radiology	Radiology role users
Pharmacy	Pharmacy role users
Admission	Ward/Admission role users
Nursing	Nursing role users
OT	Operation Theatre role users
Blood Bank	Blood Bank role users
Clinical department	Doctors linked to that department

Current limitation

Notifications are database notifications and appear after refresh. True instant notifications need WebSockets, normally Django Channels.

8. Department to Cash Counter Payment Workflow



A department can generate a pending bill from a referral. The patient takes the receipt to the Cash Counter. The cashier collects payment and changes the bill status to Paid.

Step	System behavior
Department generates bill	Bill status is Pending and linked to referral.
Receipt printed	Shows Pending, patient barcode and bill barcode.
Cashier searches/scans	Pending bills appear by patient ID, phone, or bill number.
Cashier pays selected	One or more pending bills become Paid.
Department update	Payment Completed notification is created.
Finance	Paid bills feed revenue reports.

9. Cash Counter Workflow

Cash Counter now has an active pending payment queue. Today's completed transactions remain on the dashboard. Old records are not deleted and remain searchable through reports and filters.

Requirement	Current status
Scan patient barcode	Supported through patient lookup and patient ID barcode text.
Scan bill barcode	Bill barcode encodes bill number; bill number search finds it.
Show pending bills	Pending Payments section and lookup block exist.
One-click payment	Pay Selected can mark multiple pending bills Paid.
Print receipt	Receipt available before and after payment.
Department notified	Payment completed notification sent through linked referral.

10. Department Workflow Details

Department	Current connection
Laboratory	Incoming referrals, requested tests, pending bill and payment status.
Radiology	Incoming imaging referrals and payment status.
Pharmacy	Incoming medicine/request referrals and payment workflow.
Admission	Admission referrals, ward/bed assignment, daily rate and discharge payment check.
Nursing	Can receive doctor instructions; structured nurse requests still recommended.
OT	Procedure referrals and OT billing path; richer OT booking still recommended.
Blood Bank	Blood bank referral details; structured request form still recommended.
Finance	Paid bills are available for central revenue reporting.

11. Admission and Discharge Workflow

Doctors do not admit patients directly. They send an Admission Referral. Admission staff review, admit, assign ward and bed, and manage discharge.

Admission function	Status
Patient search	Hospital ID, name, phone and scanner workflow available.
Bed availability	Available shown green, occupied shown red.
Daily ward rate	Ward.daily_rate added.
Charge estimate	Length of stay multiplied by daily ward rate.
Admission revenue	Initial metrics from paid IPD bills.
Discharge block	Discharge is blocked if patient has pending bills.
Discharge package	Recommended next step.

12. Print and Receipt Standards

Print documents should be the only area where design improvements are allowed. The current print workflow includes hospital identity, patient information, barcode and status in major documents.

Document	Print standard
Billing receipt	Hospital identity, patient barcode, bill barcode, status.
OPD ticket	Patient phone, patient barcode/QR, visit barcode, payment status.
Admission slip	Hospital identity, patient barcode, admission barcode, charge estimate.
Lab/Radiology report	Hospital identity, patient barcode, module barcode.
Pharmacy receipt	Patient barcode when patient exists, sale barcode.
OT/Blood/Insurance	Patient identity and module barcode where applicable.

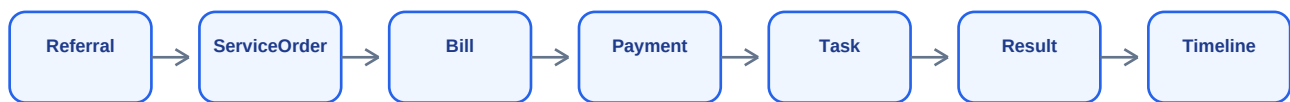
Recommendation

Create a single reusable print partial so every future document automatically uses the same header, barcode block and footer.

13. Missing Items and Recommendations

Priority	Missing or recommended	Why it matters
High	Central ServiceOrder engine	Unifies referrals, lab, radiology, pharmacy, OT, admission and blood bank orders.
High	Real-time WebSocket notifications	Departments see new requests/payment immediately without refreshing.
High	eSewa completion for pending bills	Online payment should update bill status automatically.
High	Structured Blood Bank form	Proper blood group, units, product, crossmatch and urgency fields.
High	Structured Nursing request form	Admission, transfer and discharge readiness need workflow objects.
High	Full discharge checklist	Ensures all bills, reports and documents are complete.
Medium	Pagination and filters	Required for large real hospital data.
Medium	Patient timeline	One chronological view of every patient event.
Medium	Department revenue dashboards	Departments see their own revenue; Finance sees all.
Medium	Prescription-to-sale	Doctor prescription should flow directly into pharmacy sale queue.

14. Recommended Core Architecture



The strongest next step is a central ServiceOrder model. Referrals should create one or more ServiceOrders. Each order can be billable or non-billable. Bills can cover one or many orders. Department dashboards should display orders, not scattered module-specific records.

Proposed model	Core fields
ServiceOrder	patient, referral, ordered_by, department_type, service_name, price, status, bill
PaymentEvent	bill, amount, method, received_by, paid_at, gateway_reference
DepartmentTask	service_order, assigned_role, accepted_by, completed_by, result_document
PatientTimeline	patient, event_type, source_object, timestamp, summary
Notification	role/user, title, message, related_url, read_at

15. Roadmap to Make It Best

Phase	Work	Benefit
1	Create ServiceOrder model and migrate referral billing logic.	Unified HIS backbone.
2	Build reusable department order queue for Lab, Radiology, Pharmacy, OT, Blood Bank, Nursing.	Less duplicate code and consistent workflow.
3	Add PaymentEvent and eSewa callback for pending bills.	Online payment equals cashier payment.
4	Add PatientTimeline page.	Complete chronological patient history.
5	Add WebSockets/Django Channels.	Real-time notifications.
6	Create unified print template partial.	Every print document stays consistent.
7	Add discharge checklist and final package.	Safer discharge workflow.
8	Add department revenue dashboards and analytics.	Better financial control.
9	Add audit, tests, pagination and indexes.	Production readiness and performance.

Conclusion

The system now has a working connected workflow foundation. To make it best, avoid isolated patches. Build the central order, payment, timeline and notification architecture so every department follows one clean lifecycle.